CSA1718-ARITIFICIAL INTELLIGENCE

CO3-ASSESMENT TOOL -1

CASE STUDY

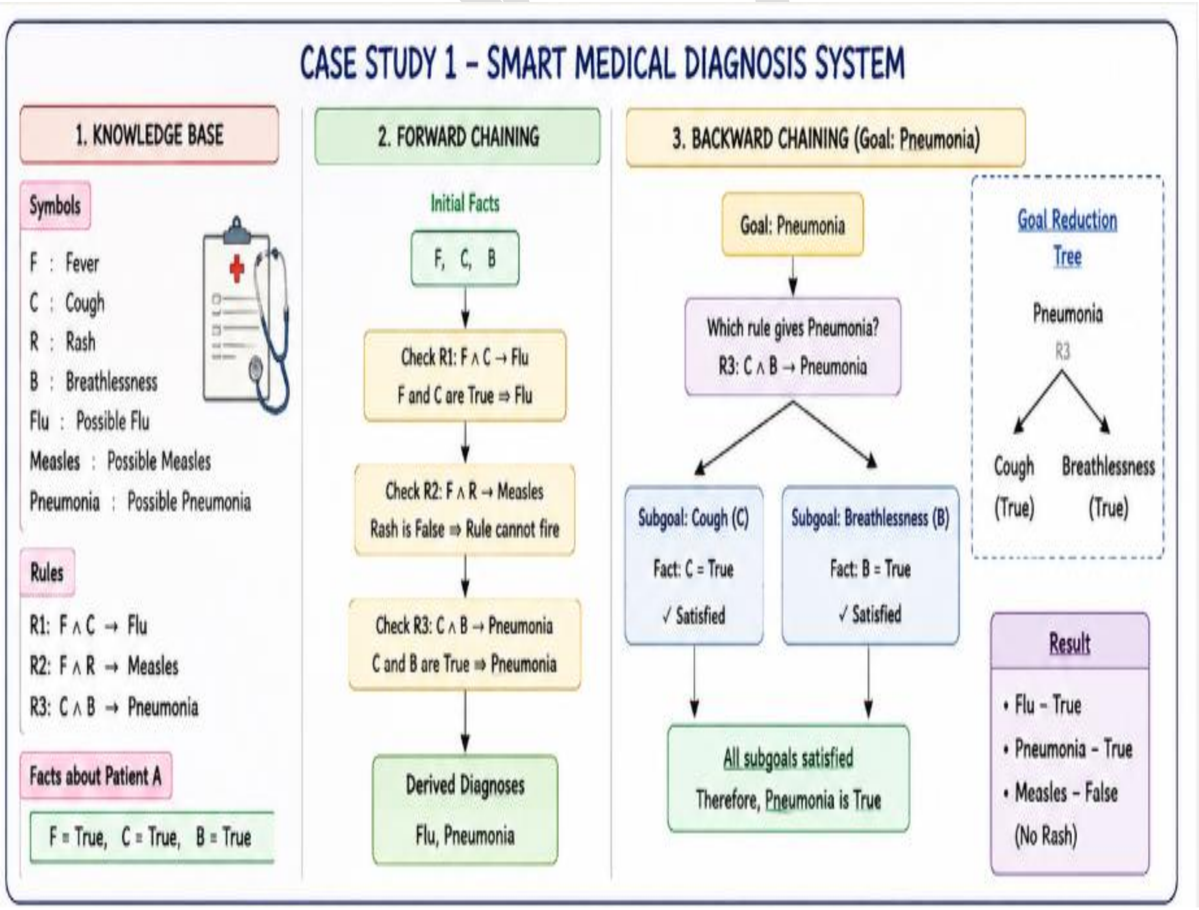
S. Harshavardhan

(192524106)

Case Study 1 – Smart Medical Diagnosis System

Introduction

A Smart Medical Diagnosis System is a rule-based logical agent that uses patient symptoms to infer possible diseases. The system applies propositional logic, forward chaining, and backward chaining to assist doctors in making preliminary diagnoses.



(a) Representation Using Propositional Logic

Define the Symbols

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Flu	Patient has Flu
Measles	Patient has Measles
Pneumonia	Patient has Pneumonia

Knowledge Base (Rules)

Rule 1

If a patient has Fever and Cough, then the patient may have Flu.

Rule 2

If a patient has Fever and Rash, then the patient may have Measles.

Rule 3

If a patient has Cough and Breathlessness, then the patient may have Pneumonia.

Facts about Patient A

Knowledge Base Summary

Facts:

- Fever
- Cough
- Breathlessness

(b) Apply Forward Chainin

Initial Facts

Fever

Cough

Breathlessness

Step 1

Check Rule 1

Fever AND Cough → Flu

Available facts:

✓ Fever

✓ Cough

Therefore,

Flu is derived.

Updated Facts

Fever

Cough

Breathlessness

Flu

Step 2

Check Rule 2

Fever AND Rash → Measles

Available facts:

✓ Fever

✗ Rash is not available

Therefore,

Rule 2 cannot be fired.

No Measles diagnosis.

Step 3

Check Rule 3

Cough AND Breathlessness → Pneumonia

Available facts:

✓ Cough

✓ Breathlessness

Therefore,

Pneumonia is derived.

Final Facts

Fever

Cough

Breathlessness

Flu

Pneumonia

Forward Chaining Inference Table

Step	Rule Fired	New Fact Derived
1	Fever ∧ Cough → Flu	Flu

2	Fever ∧ Rash → Measles	Not Fired
3	Cough ∧ Breathlessness → Pneumonia	Pneumonia

Final Diagnoses

- Flu
- Pneumonia

Measles is **not diagnosed** because Rash is absent.

(c) Apply Backward Chaining

Goal

Patient A has Pneumonia

Step 1

Goal:

Pneumonia

Look for a rule that concludes Pneumonia.

Found:

Cough AND Breathlessness → Pneumonia

Now the new goals become

- Cough
- Breathlessness

Step 2

Check Knowledge Base

Fact:

Cough = True

Goal satisfied.

Step 3

Check Knowledge Base

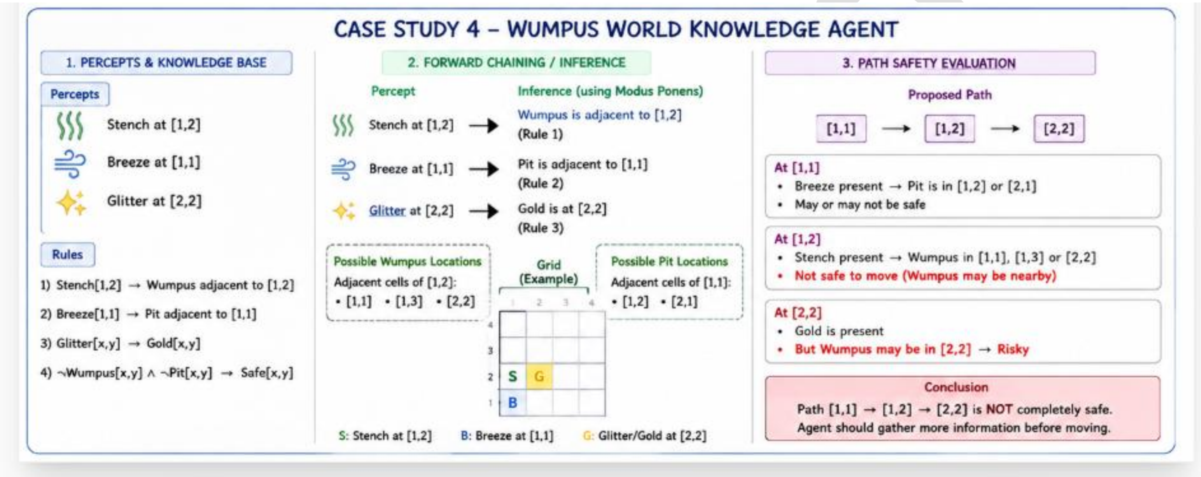
Fact:

Breathlessness = True

Case Study 4 – Wumpus World Knowledge Agent

Introduction

The Wumpus World is a well-known Artificial Intelligence environment used to demonstrate logical reasoning. An intelligent agent uses percepts such as **Stench, Breeze, and Glitter** to infer the presence of hazards (Wumpus and Pit) and locate the gold safely. The agent applies propositional logic, Modus Ponens, and Forward Chaining to make decisions.



Given Percepts

- Stench at **[1,2]**
- Breeze at **[1,1]**
- Glitter at **[2,2]**

Knowledge Base

Rule 1

Stench[1,2] → Wumpus is adjacent to cell [1,2]

Rule 2

Breeze[1,1] → Pit is adjacent to cell [1,1]

Rule 3

Glitter[x,y] → Gold[x,y]

Rule 4

¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the Agent's Belief State Using Modus Ponens

Step 1

Percept

Stench at [1,2]

Using Rule 1

Stench[1,2] → Wumpus Adjacent([1,2])

Derived Fact:

Wumpus is adjacent to cell **[1,2]**.

Step 2

Percept

Breeze at [1,1]

Using Rule 2

Breeze[1,1] → Pit Adjacent([1,1])

Derived Fact:

Pit is adjacent to **[1,1]**.

Step 3

Percept

Glitter at [2,2]

Using Rule 3

Glitter[2,2] → Gold[2,2]

Derived Fact:

Gold is present at **[2,2]**.

Belief State

Percept	Derived Knowledge
Stench at [1,2]	Wumpus is adjacent to [1,2]

Breeze at [1,1]	Pit is adjacent to [1,1]
Glitter at [2,2]	Gold is at [2,2]

Derived Conclusions

- Wumpus is near cell **[1,2]**
- Pit is near cell **[1,1]**
- Gold is located at **[2,2]**

(b) Apply Forward Chaining

Initial Facts

- Stench[1,2]
- Breeze[1,1]
- Glitter[2,2]

Rule 1 Fires

Stench[1,2]

↓

Wumpus Adjacent([1,2])

Rule 2 Fires

Breeze[1,1]

↓

Pit Adjacent([1,1])

Rule 3 Fires

Glitter[2,2]

↓

Gold([2,2])

Forward Chaining Table

Step	Rule Applied	New Fact
1	Stench → Wumpus Adjacent	Wumpus adjacent to [1,2]
2	Breeze → Pit Adjacent	Pit adjacent to [1,1]
3	Glitter → Gold	Gold at [2,2]

Possible Locations

Wumpus

Adjacent cells of **[1,2]** are

- [1,1]
- [1,3]
- [2,2]

Possible Wumpus locations:

- [1,1]
- [1,3]
- [2,2]

Pit

Adjacent cells of **[1,1]** are

- [1,2]
- [2,1]

Possible Pit locations:

- [1,2]
- [2,1]

(c) Is the Path Safe?

Path

[1,1] → [1,2] → [2,2]

Analysis

At **[1,2]**

Stench is detected.

This indicates the Wumpus is in one of the neighboring cells.

Therefore,

Moving to **[1,2]** is **not guaranteed to be safe**.

At **[2,2]**

Gold is present.

However,

Cell **[2,2]** is also a **possible location of the Wumpus**.

Hence, the agent cannot conclude that this cell is safe.

Final Decision

The path

[1,1] → [1,2] → [2,2]

is NOT completely safe because:

- Stench indicates a nearby Wumpus.
- The exact Wumpus location is unknown.
- Cell [2,2] may contain the Wumpus even though it has the gold.

The agent should collect additional percepts before moving.

Conclusion

The Wumpus World agent successfully uses **Modus Ponens** and **Forward Chaining** to infer the locations of the Wumpus, Pit, and Gold. The belief state indicates that the gold is at **[2,2]**, but the presence of a stench near **[1,2]** means the Wumpus may also be nearby. Therefore, the proposed path **[1,1] → [1,2] → [2,2]** cannot be considered completely safe without additional information. This demonstrates how logical reasoning helps AI agents make informed decisions in uncertain environments.